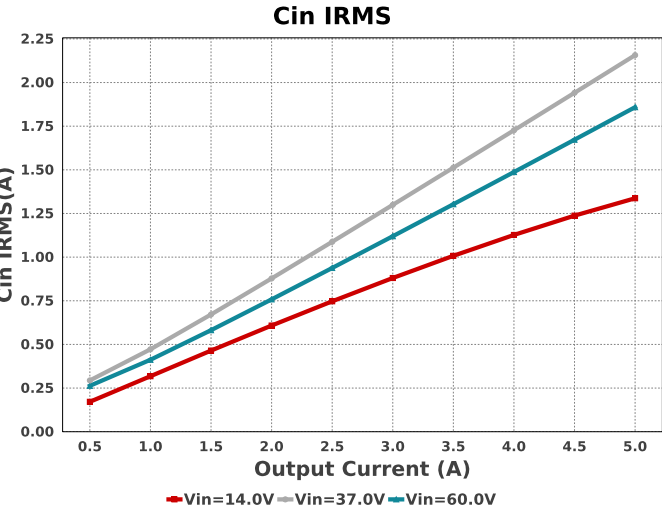
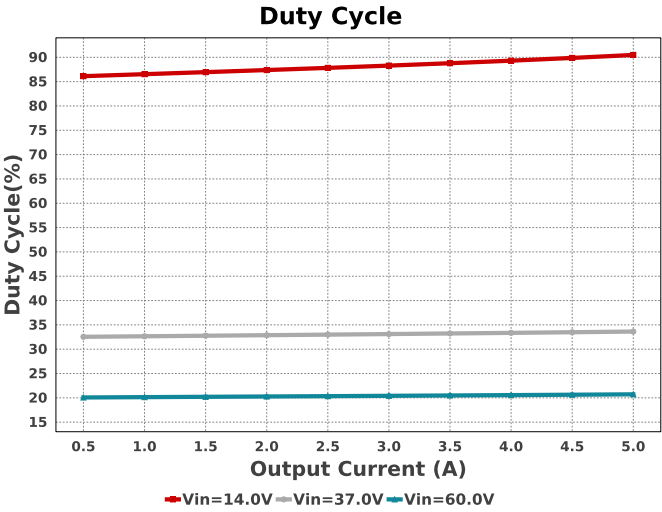
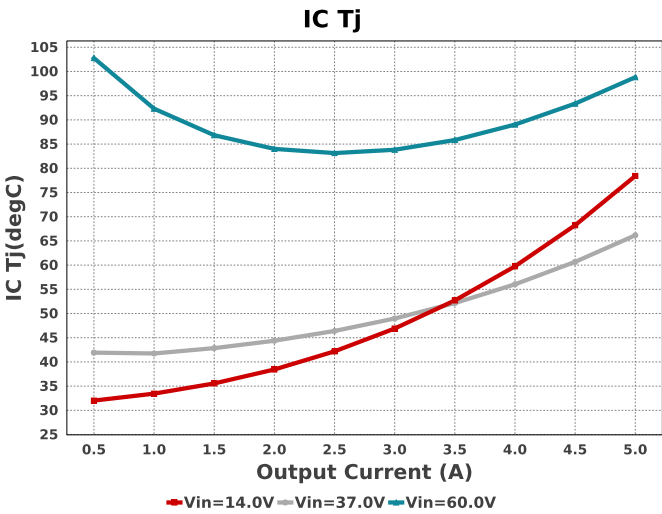
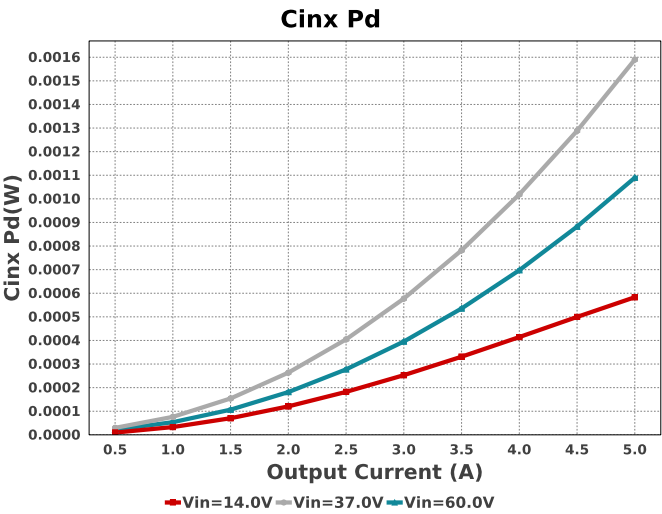
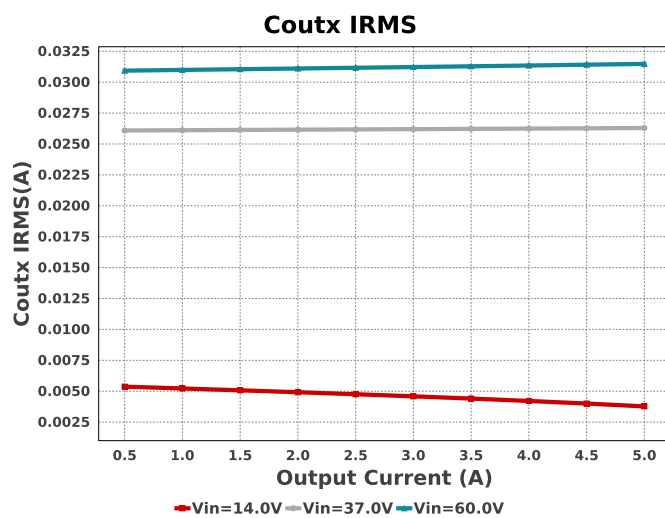
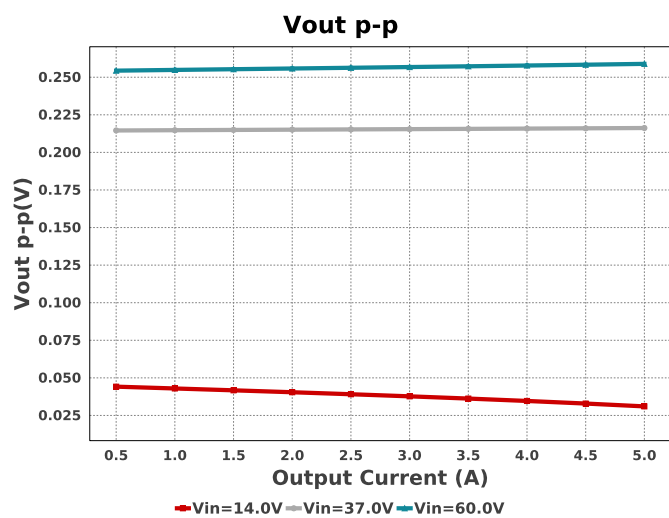
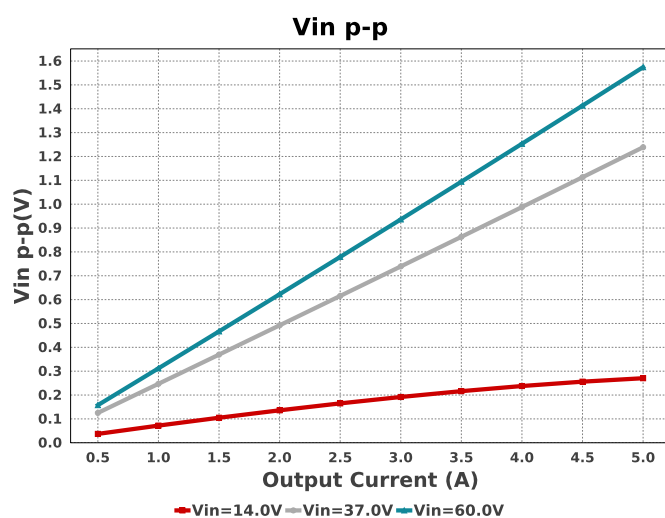
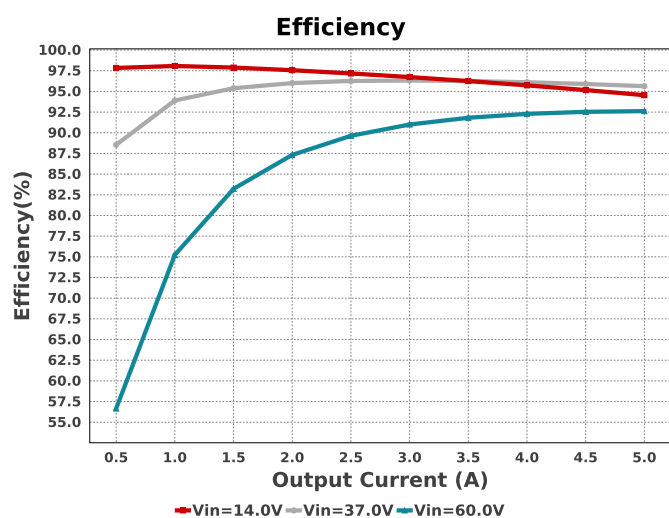
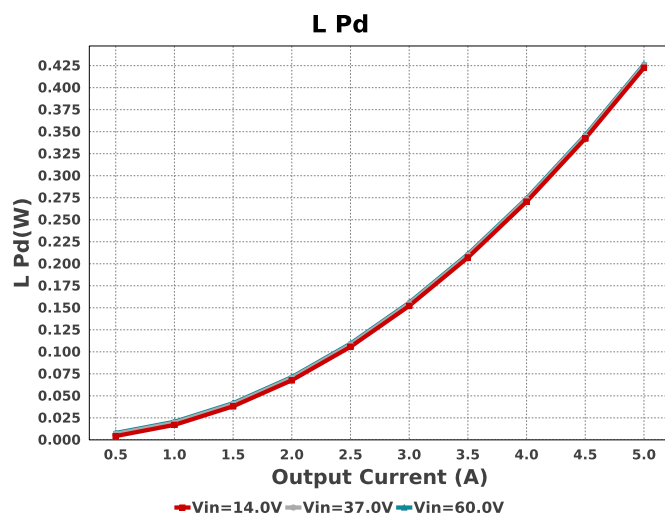
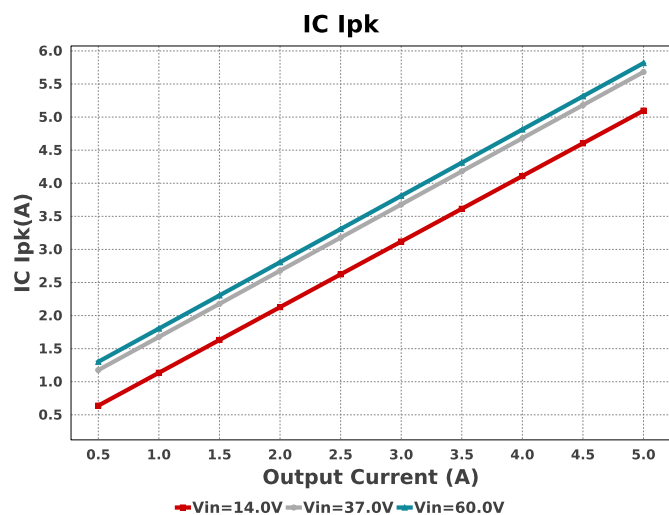


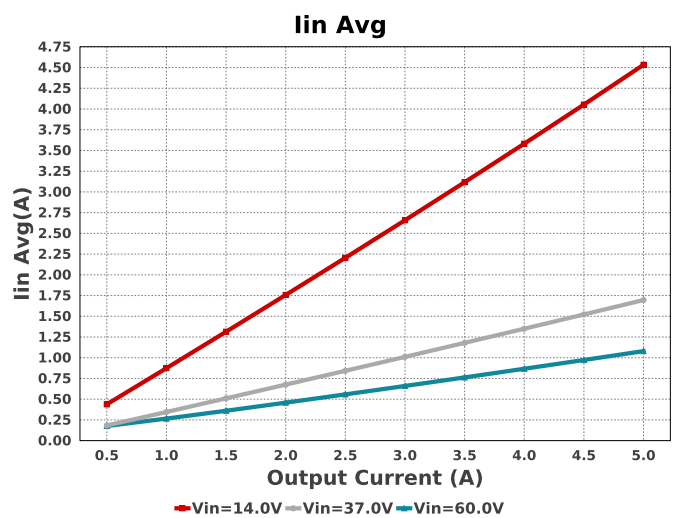
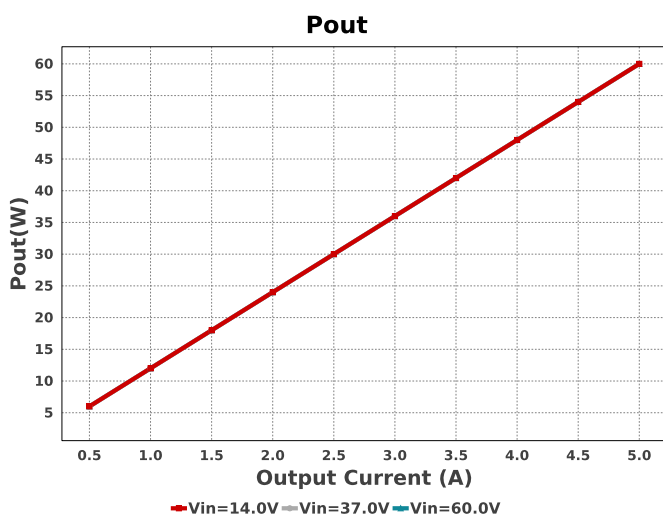
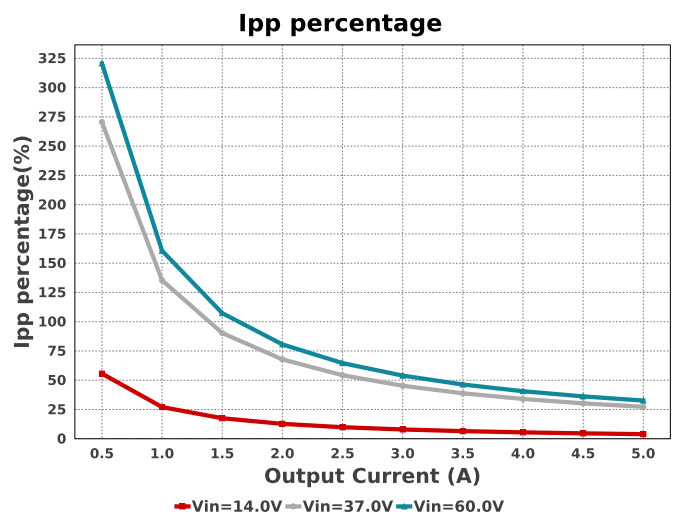
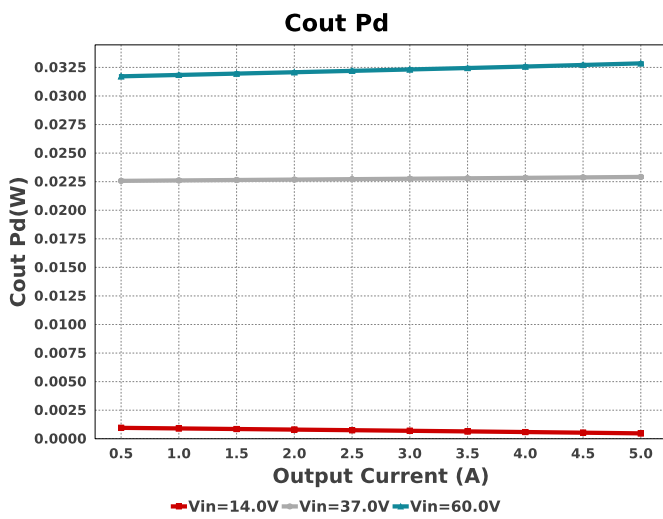
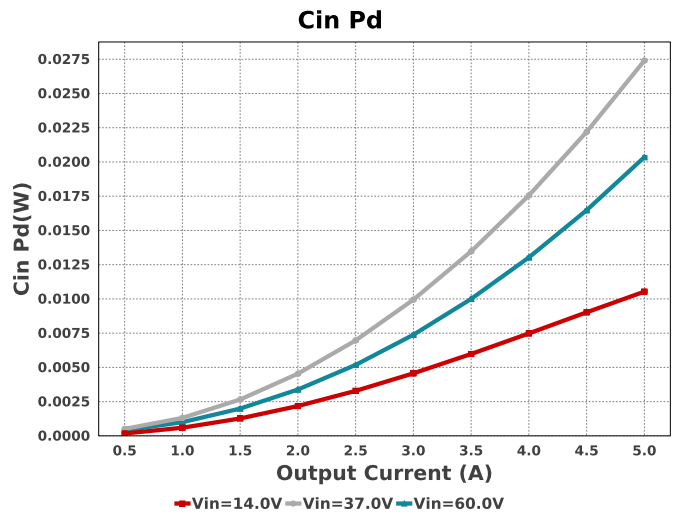
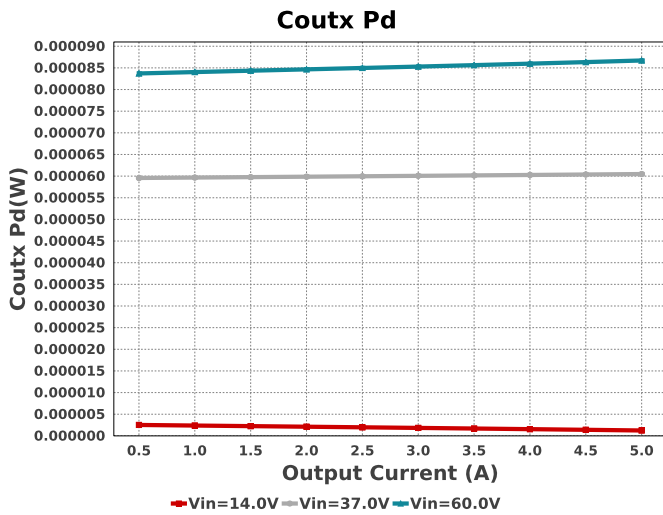


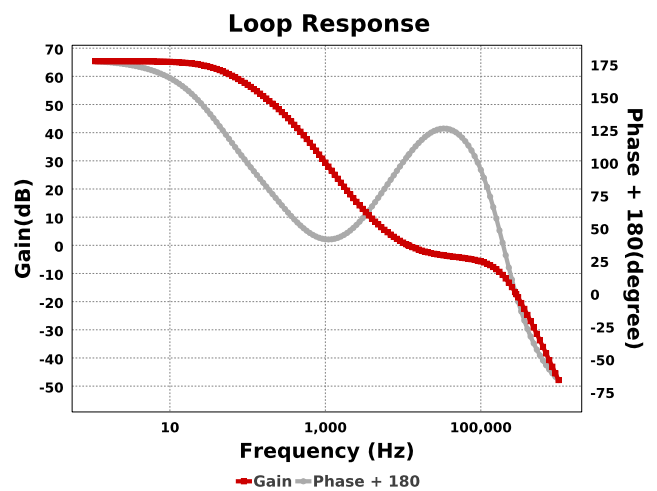
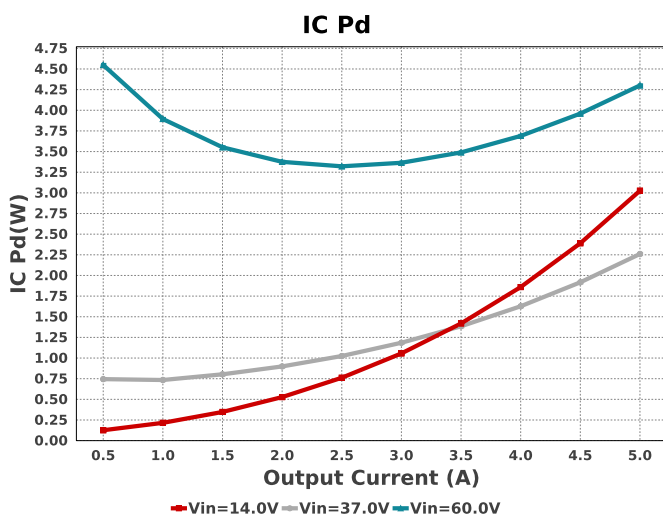
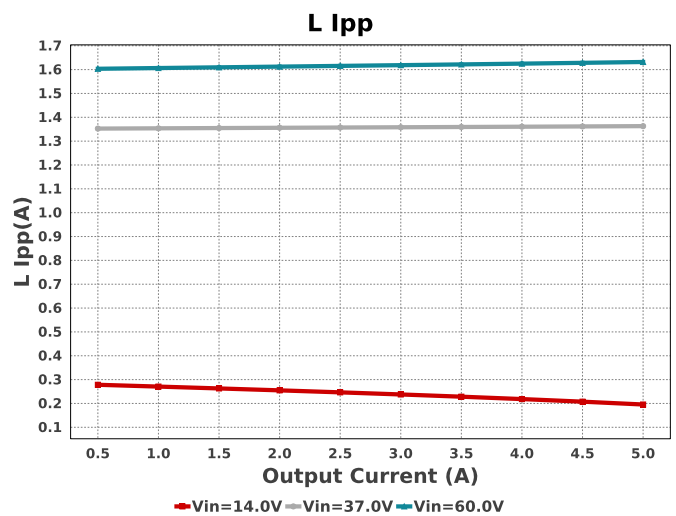
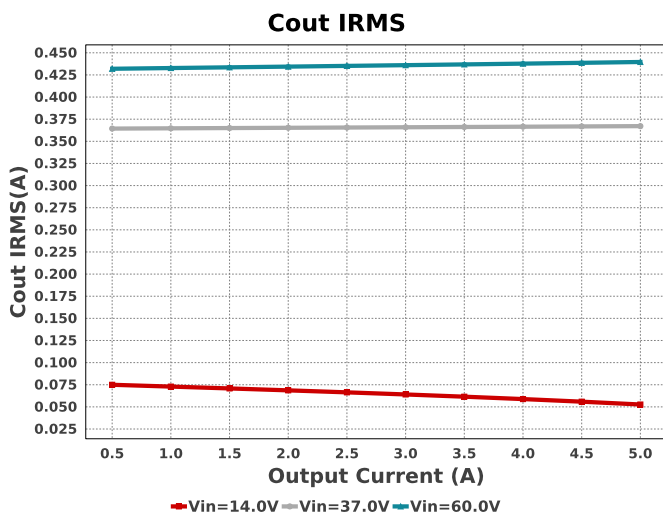
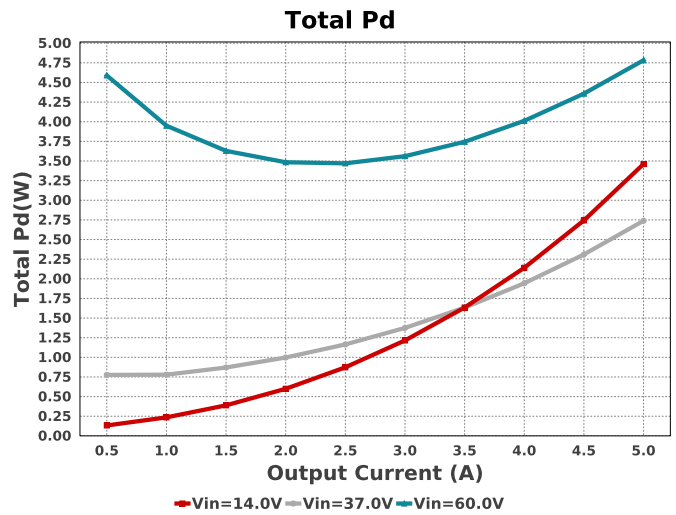
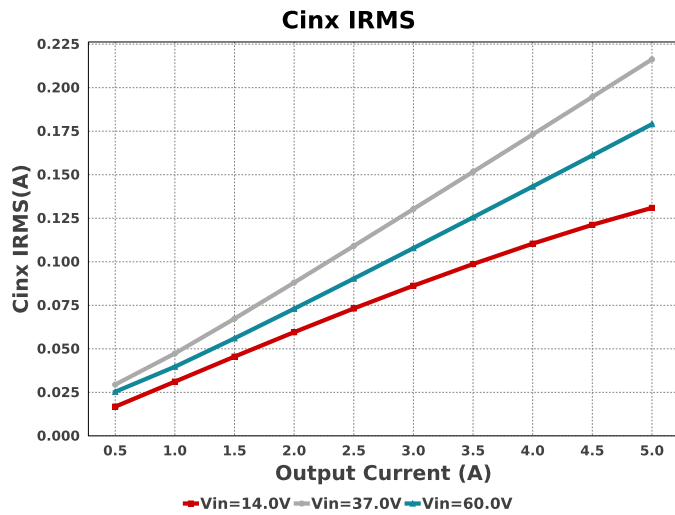
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Topology = Buck
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BOM Cost = \$5.17
BOM Count = 15
Total Pd = 4.78W

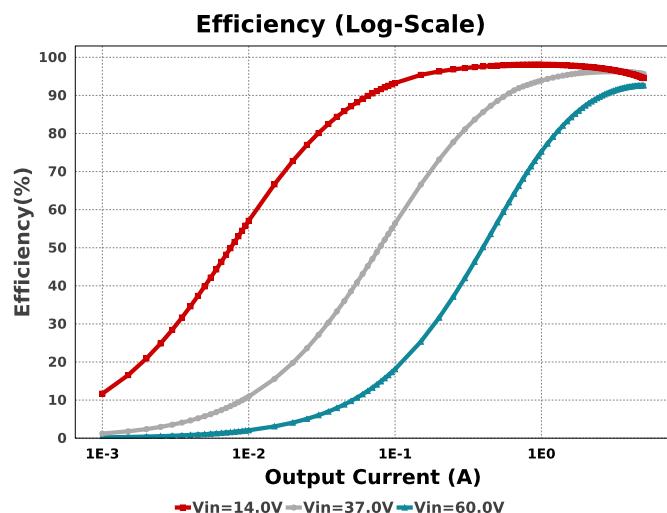
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Renb	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm²
Rent	Panasonic	ERJ-6ENF9533V Series= ERJ-6E	Res= 953.0 kOhm Power= 125.0 mW Tolerance= 1.0%	1	\$0.01	0805 7 mm²
Rfbb	Yageo	RC0603FR-079K09L Series= ?	Res= 9.09 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm²
Rfbt	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm²
Rpgood	Vishay-Dale	CRCW0603100KFKEA Series= CRCW..e3	Res= 100.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm²
U1	Texas Instruments	LM76005QRNPRQ1	Switcher	1	\$2.47	RNP0030B 48 mm²











Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	1.859 A	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	20.351 mW	Capacitor	Input capacitor power dissipation
3.	Cinx IRMS	179.031 mA	Capacitor	Bulk capacitor RMS ripple current
4.	Cinx Pd	1.09 mW	Capacitor	Bulk capacitor power dissipation
5.	Cout IRMS	439.619 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	32.855 mW	Capacitor	Output capacitor power dissipation
7.	Coutx IRMS	31.481 mA	Capacitor	Output capacitor_x RMS ripple current
8.	Coutx Pd	86.711 μ W	Capacitor	Output capacitor_x power loss
9.	IC Ipk	5.816 A	IC	Peak switch current in IC
10.	IC Pd	4.301 W	IC	IC power dissipation
11.	IC Tj	98.811 degC	IC	IC junction temperature
12.	IC Tolerance	20.0 mV	IC	IC Feedback Tolerance
13.	ICThetaJA Effective	16.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
14.	Iin Avg	1.08 A	IC	Average input current
15.	Ipp percentage	32.639 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
16.	L Ipp	1.632 A	Inductor	Peak-to-peak inductor ripple current
17.	L Pd	426.25 mW	Inductor	Inductor power dissipation
18.	Cin Pd	20.351 mW	Power	Input capacitor power dissipation
19.	Cinx Pd	1.09 mW	Power	Bulk capacitor power dissipation
20.	Cout Pd	32.855 mW	Power	Output capacitor power dissipation
21.	Coutx Pd	86.711 μ W	Power	Output capacitor_x power loss
22.	IC Pd	4.301 W	Power	IC power dissipation
23.	L Pd	426.25 mW	Power	Inductor power dissipation
24.	Total Pd	4.783 W	Power	Total Power Dissipation
25.	BOM Count	15	System	Total Design BOM count
26.	Cross Freq	11.719 kHz	System	Bode plot crossover frequency
27.	Duty Cycle	20.72 %	System	Duty cycle
28.	Efficiency	92.617 %	System	Steady state efficiency
29.	FootPrint	418.0 mm ²	System	Total Foot Print Area of BOM components
30.	Frequency	400.0 kHz	System	Switching frequency
31.	Gain Marg	-17.329 dB	System	Bode Plot Gain Margin
32.	Iout	5.0 A	System	Iout operating point
33.	Low Freq Gain	65.375 dB	System	Gain at 1Hz
34.	Mode	FCCM	System	Conduction Mode
35.	Phase Marg	106.79 deg	System	Bode Plot Phase Margin
36.	Pout	60.0 W	System	Total output power
37.	Total BOM	\$5.17	System	Total BOM Cost

#	Name	Value	Category	Description
38.	Vin	60.0 V	System Information	Vin operating point
39.	Vin p-p	1.574 V	System Information	Peak-to-peak input voltage
40.	Vout	12.0 V	System Information	Operational Output Voltage
41.	Vout Actual	12.001 V	System Information	Vout Actual calculated based on selected voltage divider resistors
42.	Vout Tolerance	3.889 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
43.	Vout p-p	258.891 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	5.0	Maximum Output Current
SoftStart	15.0 ms	Soft Start Time (ms)
VinMax	60.0	Maximum input voltage
VinMin	14.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	LM76005-Q1	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature
UserFsw	400.0 k	Customer Selected Frequency

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

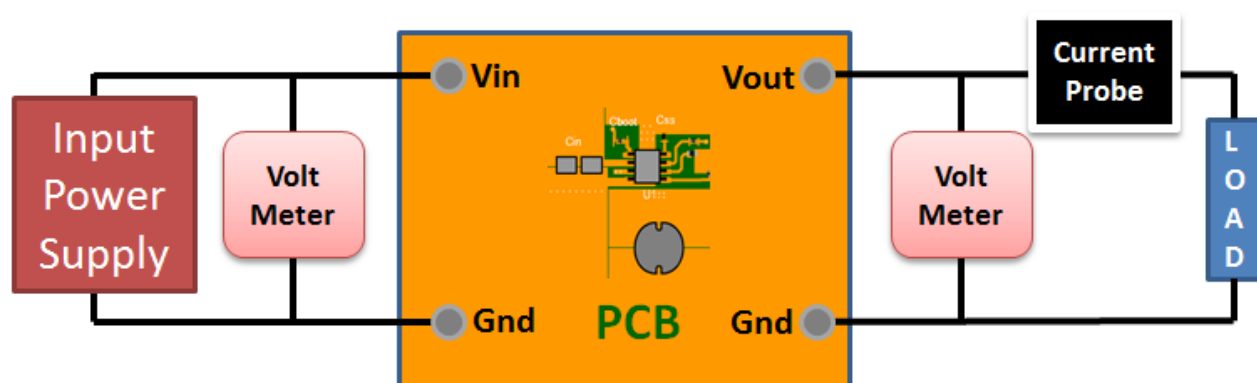
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 14.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. The LM76005-Q1 is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application
2. Master key : 3A09369785B968091A003B956300BD89[v1]
3. **LM76005-Q1** Product Folder : <http://www.ti.com/product/LM76005%2DQ1> : contains the data sheet and other resources.

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